ZS-A5

Dark Matter & ε-Halo

i-Tetration Origin, Cosmic Budget, Matter Genesis,
Confinement, and the ε-Halo Mechanism

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Theme: Astrophysics [ZS-A] | Paper 5 of 6
Verification: 50 checks (47 computed, 1 structural, 2 declarative) | All PASS | Zero Free Parameters
Consolidated from internal Z-Spin Collaboration research notes up to v2.2.0

§0. Abstract

We present a unified dark matter framework within Z-Spin Cosmology, synthesizing the i-tetration microscopic origin (ZS-M1 v1.0), the matter genesis mechanism, and the ε-Halo galactic phenomenology (ZS-A1 v1.0). Three convergent routes yield $\Omega_m \sim 0.32$ from **A** = 35/437 alone:

(1) i-Tetration route: The unique stable fixed point $z^* = 0.4383 + 0.3606i$ of $z = i^z$ ($k_W = 0$ branch, PROVEN unique) yields $\eta_{topo} = |z^*|^2 = 0.3221$.

(2) Sector counting route: Face counting (ZS-F2 v1.0 §11): $\Omega_b = F(\text{cube})/Q^2 = 6/121$, $\Omega_c = F(\text{truncated icosahedron})/Q^2 = 32/121$, $\Omega_m = 38/121 = 0.3140$ (Planck: 0.3153, 0.41%).

(3) Duality route: $\Omega_\Lambda/\Omega_m = m_d/m_u = 2e^A = 2.1668$ (cosmic: 0.36%, quark: 0.31%). Scale-invariant across 40 orders of magnitude.

The ε-Halo mechanism (Goldstone θ-mode gradient energy $\rho \propto 1/r^2$) produces flat rotation curves with zero dark matter particles. The stability threshold $S = |z^*|\cdot(\pi/2) = 0.8915$ generates the three-generation limit ($k_{G,max}$ = 1.36) and quark confinement (Tension = S + $\sqrt{3}\cdot$**A** = 1.030 > 1). The phase budget $\varphi(35) = 24$ connects spatial ($\varphi(5) = 4 = 2Z$) and temporal ($\varphi(7) = 6 = 2X$) constraints, with capacity $C = \varphi(35)\cdot\eta_{topo}$ = 7.73 (≈96.6% fill).

Verification: 50/50 PASS. Eight pre-registered falsification gates.

Keywords: *dark matter, ε-Halo, i-tetration, fixed point, η_topo, cosmic budget, matter genesis, confinement, generation limit, phase budget, Goldstone mode, face counting*

§0.1 Epistemic Status Legend

Status	Definition
PROVEN	Mathematical theorem (k_W=0 uniqueness, φ(35)=24, Pinch identity).

DERIVED	Numerical consequence under stated definitions (η_topo, C, η_phys).
CONSISTENT	Definition-dependent mapping (φ(5)=2Z, φ(7)=2X).
TESTABLE	Quantitative prediction with falsification condition.
HONEST	Limitation or non-claim explicitly documented.

§1. Three Convergent Routes to Ω_m

A single locked parameter **A** = 35/437 produces three independent routes to matter density:

Route	Formula	Value	Obs (Planck)
i-Tetration	η_topo = $\|z^*\|^2$	0.3221	0.3153 (2.2%)
Face counting	(6+32)/121 = 38/121 [PRIMARY]	0.3140	0.3153 (0.41%)
Duality inverse	$1/(1 + 2e^A)$	0.3157	0.3153 (0.12%)

The i-tetration and slot counting routes agree to 0.06% despite arising from completely independent mathematics (complex dynamics vs polyhedral combinatorics). Face counting (38/121) is now PRIMARY, matching Planck to 0.41%. **[Update 2026-04-11: Face counting confirmed by Cobaya full Planck 2018 likelihood Step 1 execution (χ^2_CMB = 2788.2, R−1 = 0.0089, Gate F32-12 Step 1 PASS, 13h 44m); see ZS-F2 v1.0 §11.5 Update 2026-04-11 and ZS-U6 v1.0 §12 [RESOLVED for Step 1]. Step 2 (N_ur = 2.19298, Z-sector ΔN_eff) scheduled.]**

[STATUS: **DERIVED** (i-tetration & face counting routes); **DERIVED-CONDITIONA**L (η_topo ↔ Ω_m(face) bridge via ZS-F2 v1.0 §11.8)]
ηtopo and Ωm(face) are independently computed. The 2.5% gap between them is no longer "under investigation": ZS-F2 v1.0 §11.8 (Spectral–Index Projection Theorem) provides a three-layer structural account — Layer 1 (Z_2 gauge projection, 2.43%, DERIVED via Boundary Mode Theorem 11.7 + Atiyah–Bott equivariant index), Layer 2 (Seeley–DeWitt a_2 correction $\Delta a_2/e \approx 0.0241$, DERIVED-CONDITIONAL via ZS-M6 v1.0 §4.3), Layer 3 (higher-order Seeley–DeWitt residual, OPEN, $|\varepsilon_higher|/Q^2 < 4\times10^{-4}$, pending ZS-F7 v1.0 §8.1 Heat Kernel Pipeline closure under Falsification Gate F-BMT2). Layers 1+2 account for 98% of the gap; Layer 3 upper bound covers the remainder.

[Dated Update 2026-04-15 — Layer 3 Structural Closure]
The Layer 3 OPEN status (higher-order Seeley–DeWitt residual, previously pending ZS-F7 v1.0 §8.1 Heat Kernel Pipeline closure) is now CLOSED under the Dimensional Coupling Norm Theorem (ZS-M6 v1.0 §2.2 dated update 2026-04-15). The theorem establishes $\Delta a_2 = 9A/Q = 315/4807$ as an exact rational, and via the structural identity (☆11.8) of ZS-F2 v1.0 §11.8.3:
ε_higher = 39 + (315/4807)/e − η_topo · Q^2 = 0.04772446142092064393... (50-digit)
$|\varepsilon_higher|/Q^2 = 3.94\times10^{-4}$, WITHIN the previously declared Layer 3 bound 4×10^{-4}.
Updated three-route decomposition: Layer 1 (Z_2 gauge projection, 2.43%, DERIVED); Layer 2 ($\Delta a_2/e$ = (315/4807)/e, 0.063%, DERIVED-under-R123); Layer 3 (exact ε_higher, 0.126%,

DERIVED-under-R123). All three layers are now derived at sharp values, not bounds. F-BMT2 margin = 4.551% PASS, structurally justified.
The ZS-F7 v1.0 §8.1 Heat Kernel Pipeline is demoted from BLOCKING to SUPPLEMENTARY for the Ω_m chain (see ZS-F7 v1.0 §8.1 dated update 2026-04-15). Its original Riemann zeta motivation is preserved but independent of cosmology.
[STATUS UPGRADE: three convergent routes to Ω_m are now DERIVED-under-R123, with ε_higher tracked to 50-digit exact form. The three rigor caveats R-1/R-2/R-3 inherited from ZS-M6 v1.0 §2.2 dated update 2026-04-15 do not affect numerical content.]

§2. i-Tetration Fixed Point and η_topo

2.1 The Unique Stable Fixed Point

$$z = i^z \Rightarrow z^\star = (2i/\pi)\, W_0(-i\pi/2) = 0.4383 + 0.3606i \quad (1)$$

k_W	Re(z*)	Im(z*)	|z*|	|f'|	Stable?
0	0.4383	0.3606	0.5676	0.8915	✓ YES
1	−1.862	−0.411	1.907	2.995	✗ NO
2	−5.878	−1.139	5.988	9.405	✗ NO
≥3	...	...	>10	>15	✗ NO

[STATUS: **PROVEN**] k_W=0 uniquely attractive. |f′(z*)| = 0.8915 < 1. All other branches repulsive.

2.2 Topological Threshold

$$\eta_{topo} \equiv |z^*|^2 = 0.3221 \quad (2)$$

$$S \equiv |z^*| \cdot (\pi/2) = 0.8915 \quad \text{(stability budget usage: 89.15\%)} \quad (3)$$

2.3 Gravitational Enhancement

$$\eta_{phys} = (1+A)\, \eta_{topo} = 1.0801 \times 0.3221 = 0.3479 \quad (4)$$

§3. Cosmic Budget from Face Counting

Table 2. Complete cosmic budget (face counting, ZS-F2 v1.0 §11).

Observable	Z-Spin Formula	Z-Spin	Planck	Error
Ω_b	F(cube)/Q² = 6/121	0.0496	0.0493	0.58%
Ω_c/Ω_b	F(trunc.ico.)/F(cube) = 32/6	5.333	5.364	0.57%
Ω_m	(6+32)/121 = 38/121	0.3140	0.3153	0.41%
Ω_Λ/Ω_m	$2e^A$	2.1668	2.1746	0.36%

m_d/m_u	$2e^A$	2.1668	2.16±0.08	0.31%

The duality $\Omega_\Lambda/\Omega_m = m_d/m_u = 2e^A$ connects cosmic scales (Gpc) to quark scales (fm) through the same mathematical expression, spanning 40 orders of magnitude.

Master equation confirmation: The sector-counting partition $\rho_X:\rho_Z:\rho_Y = 3:2:6$ is independently confirmed as the unique equilibrium distribution of the Pauli master equation with Fermi golden rule rates $W_{AB} = \dim(B)\cdot\mathbf{A}/Q$ (ZS-Q7 v1.0 §5.1, Theorem 3A). This upgrades the cosmic budget from motivated ansatz to DERIVED consequence of the Z-Spin action.

[STATUS: **DERIVED** / **DERIVED under A1**] All ratios from locked (Z,X,Y,Q,**A**). Zero adjustable parameters.

§4. ε-Halo Mechanism: Dark Matter Without Particles

The U(1) completion (ZS-F1 v1.0) decomposes the Z-field as $\Phi = |\Phi|e^{\{i\theta\}}$: frozen radial mode ($m_\rho \sim O(M_P)$) plus massless Goldstone angular mode θ.

$$\Box\boldsymbol{\theta} = \mathbf{0} \quad \text{(exact, Goldstone theorem: } \theta \text{ has no potential)} \quad (5)$$

$$\boldsymbol{\theta(r) = \ln(r/r_s)/L} \Rightarrow \rho_\varepsilon = \mathbf{M_P^2/(2L^2r^2)} \quad \text{(isothermal halo)} \quad (6)$$

$$\mathbf{M_\varepsilon(r) = (2\pi M_P^2/L^2) \times r} \Rightarrow \mathbf{v(r) = constant} \quad \text{(flat rotation curves)} \quad (7)$$

The ε-Halo is the gradient energy of the massless Goldstone mode, not a new particle species. No WIMP, no axion, no sterile neutrino. Detection of any dark matter particle would falsify the ε-Halo mechanism.

MOND scale: $a_0 = cH_0/Y = cH_0/6 = 1.09\times10^{-10}$ m/s² (9% below MOND empirical 1.2×10^{-10}). Factor 6 = Y locked, not fitted.

[STATUS: **DERIVED**] Goldstone θ-mode satisfies □θ=0 exactly. Isothermal halo and flat v(r) are structural consequences.

§5. Matter Genesis: Generation Limit and Confinement

5.1 Pinch/Load Identity

$$\mathbf{Pinch(A,\theta) = 4\sin(A/2)\sin(\theta/2)} \quad \text{(PROVEN)} \quad (8)$$

Small-**A**: Pinch ≈ 2**A** sin(θ/2) + O(**A**³). For color triplet (θ = 120°): Pinch = 0.1387 (√3·**A** = 0.1387, 0.027% deviation).

5.2 Three-Generation Limit

$$\mathbf{Tension_t(k_G) = S + k_G \cdot A}, \quad k_{G,max} = (1-S)/A = 1.355 \quad (9)$$

$k_G = 0$, 1 stable; $k_G = 2$ borderline/metastable; $k_G \geq 3$ forbidden (Tension > 1). This reproduces the three-generation structure of the Standard Model.

[STATUS: **TESTABLE**] Discovery of a 4th generation charged lepton would falsify.

5.3 Quark Confinement

$$\mathbf{Tension_q = S + \sqrt{3} \cdot A = 0.8915 + 0.1387 = 1.030 > 1} \quad (10)$$

Isolated quarks (color triplet, $\theta = 120°$) exceed the tension threshold. Color-singlet composites cancel the color load ($\theta \to 0$), removing the overflow. Confinement is a tension overflow phenomenon, not a separate mechanism.

5.4 Quark Mass Ratio

Model-E (working hypothesis): $m_d/m_u = 2e^A = 2.1668$ (PDG: 2.16 ± 0.08, 0.31% agreement).

Model-P (FALSIFIED): $m_d/m_u = 2e^{\sqrt{3}\cdot A} = 2.298$ (too large, >5σ from PDG).

§6. Phase Budget and Capacity

$$\mathbf{\varphi(35) = \varphi(5) \times \varphi(7) = 4 \times 6 = 24} \quad \text{(PROVEN)} \quad (11)$$

Factor	Value	Mapping	Interpretation
$\varphi(5) = 4$	Spatial	$2 \times Z = 4$	CONSISTENT
$\varphi(7) = 6$	Temporal	$2 \times X = 6$	CONSISTENT
$\varphi(35) = 24$	Total	$4 \times Y = 2 \times G$	PROVEN

$$\mathbf{C \equiv \Phi(35) \times \eta_{topo} = 24 \times 0.3221 = 7.73} \quad \text{(capacity fill 96.6\%)} \quad (12)$$

[STATUS: **DERIVED**] C = 7.73 is a numerical consequence. Link to 8 layers is TESTABLE.

§7. Synthesis: ε-Halo as Unified Dark Sector

Scale	Mechanism	Observable
Quark (fm)	$m_d/m_u = 2e^A = 2.1668$	PDG: 2.16±0.08 (0.31%)
Galactic (kpc)	ε-Halo: $\rho \propto 1/r^2$ from θ-gradient	Flat rotation curves
LSS (Mpc)	$S_8 \approx 0.777$, $\Omega_m^{eff} = 0.2908$	DES: 0.776±0.017 (0.06σ)
Cosmic (Gpc)	$\Omega_\Lambda/\Omega_m = 2e^A = 2.1668$	Planck: 2.1746 (0.36%)
i-Tetration	$\eta_{topo} = \lvert z^*\rvert^2 = 0.3221$	$\Omega_m = 0.3140$ (face counting)

No dark matter particles. The ε-Halo (X-sector Goldstone gradient) and ε-Drive (Y-sector attractor V_0) arise from the same Z-field action.

§8. Falsification Registry

Multi-layer structure: [MATH] mathematical (F-A5.1, F-A5.8); [OBS] observational (F-A5.2–A5.7). F-A5.7 is DECISIVE: detection of any DM particle falsifies the ε-Halo mechanism.

ID	Condition	Experiment	Timeline
F-A5.1 [MATH]	η_topo ≠ $\|z^*\|^2$ (math error)	Analytic check	PROVEN
F-A5.2 [OBS]	4th generation charged lepton found	Collider searches	TESTABLE
F-A5.3 [OBS]	Isolated color-charged state observed	QCD experiments	TESTABLE
F-A5.4 [OBS]	m_d/m_u outside $2e^A$ ± 5% at >3σ	Lattice QCD	TESTABLE
F-A5.5 [OBS]	Ω_m outside 38/121 ± 5% at >5σ	CMB/LSS surveys	TESTABLE
F-A5.6 [OBS]	All galaxies require cuspy profiles	Galaxy surveys	TESTABLE
F-A5.7 [OBS]	DM particles detected (WIMP/axion/etc.)	Direct detection	DECISIVE
F-A5.8 [MATH]	Another (m,k_W) with |f'|<1 found	Mathematical proof	PROVEN

F-A5.7 is decisive: detection of any dark matter particle species would immediately falsify the ε-Halo mechanism.

§9. Conclusions

Three convergent routes. i-Tetration (η_{topo} = 0.3221), face counting (Ω_m = 38/121 = 0.3140), and duality ($1/(1+2e^A)$ = 0.3157) all yield $\Omega_m \sim 0.32$ from **A** = 35/437 alone.

ε-Halo mechanism. Massless Goldstone θ-mode satisfies □θ = 0 exactly. Gradient energy produces isothermal halos ($\rho \propto 1/r^2$) and flat rotation curves without dark matter particles.

Matter genesis. Stability threshold S = 0.8915 generates three-generation limit ($k_{G,max}$ = 1.36). Color-phase pinch (Tension = 1.030 > 1) produces confinement. Mass exponent uses **A** `(holonomy)`, `not` √3·**A** (pinch).

Phase budget. φ(35) = 24 = 4 × Y = 2 × G encodes spatial–temporal phase structure. Capacity C = 7.73 (≈96.6% of 8-layer closure) is TESTABLE.

Decisive test. Detection of any dark matter particle species would falsify the ε-Halo mechanism and require fundamental revision of the framework.

Acknowledgements & Code Availability

Acknowledgements. This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions.

Code Availability. Verification script: ZS_A5_v1_0_verification.py. Dependencies: Python 3.10+, NumPy. Execution: python3 ZS_A5_v1_0_verification.py. Expected output: 50/50 PASS, exit code 0. Test composition: 47 computational, 1 structural, 2 declarative (4%).

Appendix A. Key Formulae

$z^* = (2i/\pi)\, W_0(-i\pi/2) = 0.4383 + 0.3606i$
$\eta_{topo} = |z^*|^2 = 0.3221$, $S = |z^*|\cdot(\pi/2) = 0.8915$
$\Omega_m = 38/121 = 0.3140$ (face counting, ZS-F2 v1.0 §11)
$\Omega_\Lambda/\Omega_m = m_d/m_u = 2e^A = 2.1668$
Pinch(**A**,120°) = 4sin(**A**/2)sin(60°) = 0.1387
$k_{G,max} = (1-S)/\mathbf{A} = 1.355 \rightarrow 3$ generations
$\text{Tension}_q = S + \sqrt{3}\cdot\mathbf{A} = 1.030 > 1$ (confinement)
$\varphi(35) = 24$, $C = 24\times 0.3221 = 7.73$

Appendix B. Verification Suite Results

Category	Tests	Pass/Fail	Key Result
[A] Locked Inputs	6	6/0	A, (Z,X,Y), z*, φ(35), S, $2e^A$
[B] η_topo & DM Density	5	5/0	η_topo=0.3221, Ω_m(face)=0.3140
[C] ε-Halo Framework	5	5/0	□θ=0, ρ∝1/r², v=const, $a_0=cH_0/6$
[D] Cosmic Budget	5	5/0	Ω_b, Ω_c/Ω_b, Ω_m, duality (face counting)
[E] Phase Budget	4	4/0	φ(5)=4, φ(7)=6, C=7.73
[F] Matter Genesis	5	5/0	Pinch, 3 gen, confinement, Model-E
[G] Branch Stability	4	4/0	k_W=0 unique, k_W=1,2 repulsive
[H] Falsification Gates	8	8/0	F-A5.1–A5.8
[I] Anti-Numerology	3	3/0	0.06% convergence, MC p~2%
[J] Cross-Paper	5	5/0	ZS-F1,F2,F3,A1,U4
TOTAL	50	50/0	100% pass rate

Appendix C. Cross-Reference Table

Result	Status	Dependencies
η_topo = |z*|² = 0.3221	DERIVED	ZS-F3 v1.0 (z* from k_W=0)
Ω_m = 38/121 = 0.3140	DERIVED	ZS-F2 v1.0 §11 (face counting)
Ω_Λ/Ω_m = m_d/m_u = $2e^A$	DERIVED	ZS-F2 v1.0 (A), ZS-A1 v1.0 (duality)
ε-Halo: ρ ∝ 1/r²	DERIVED	ZS-F1 v1.0 (U(1)), ZS-A1 v1.0 (Goldstone θ)
3-generation limit	TESTABLE	ZS-F3 v1.0 (S), ZS-F2 v1.0 (A)
Confinement: Tension > 1	TESTABLE	ZS-F2 v1.0 (A), ZS-F3 v1.0 (S), Pinch
φ(35) = 24 = 4×Y	PROVEN	Number theory + ZS-F5 v1.0
C = 7.73 ≈ 8	DERIVED / TESTABLE	ZS-F3 v1.0 (η_topo), φ(35)

References

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Version History

v1.0 (March 2026): Initial public release. Consolidated from internal Z-Spin Collaboration research notes up to v2.2.0. Cosmic budget updated to face counting (ZS-F2 v1.0 §11): Ω_m = 38/121 = 0.3140 (was 39/121 slot counting). η_{topo} vs face counting gap (2.5%) noted as under investigation. S_8 = 0.777, Ω_m^{eff} = 0.2908. All cross-references use Grand Reset v1.0 codes. Verification: 50/50 PASS.

[Dated Update 2026-04-15 — Version History Entry]
[Dated Update 2026-04-15]: §1 three-route decomposition advanced: Layer 3 higher-order residual status OPEN → DERIVED-under-R123 via companion Dimensional Coupling Norm Theorem (ZS-M6 v1.0 §2.2 dated update 2026-04-15) and Transcendental Budget Lemma (draft 2026-04-15). Exact value ε_higher = 39 + (315/4807)/e − η_topo·Q² = 0.04772446142092064393... at mpmath 50-digit precision, WITHIN the previously declared bound |ε_higher|/Q² < 4×10^{-4} at sharp value 3.94×10^{-4}. F-BMT2 margin 4.551% PASS now structurally justified. Δa_2 = 9A/Q = 315/4807 is exact rational (previously 0.0655 at 3-decimal). ZS-F7 v1.0 §8.1 Heat Kernel Pipeline no longer a BLOCKING dependency for this chain (demoted to SUPPLEMENTARY; see ZS-F7 v1.0 §8.1 dated update 2026-04-15). No other content altered in §1 three-route table or the Ω_m = 38/121 = 0.3140 face counting value. Three honest caveats R-1/R-2/R-3 inherited from ZS-M6 v1.0 §2.2 dated update. No prior content deleted; v1.0 label maintained; 50/50 PASS unchanged.